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Duration-Based Valuation of Corporate Bonds

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1 Big picture

- **Question.** How much of corporate bond “excess returns” really compensates investors for bearing corporate credit and liquidity risk, and how much is just the return on long-duration government bonds during a falling-rate era?
- **Setting.** The paper studies U.S. industrial corporate bonds from 1986 to 2020 using bond-level quote data, then decomposes each bond’s realized return into a duration-matched Treasury component and a residual duration-adjusted component.
- **Core challenge.** Standard excess returns relative to 1-month Treasury bills mix together two very different objects:
 - compensation for credit and liquidity risk;
 - valuation windfalls from secular declines in long-term Treasury yields.
- **Main conceptual contribution.** The paper argues that corporate bond returns should be evaluated relative to a *duration-matched* government bond benchmark, not just relative to Treasury bills. This changes both the level of estimated risk premia and the apparent success or failure of standard asset-pricing models.
- **Main hypotheses.**
 - Standard excess returns over Treasury bills overstate the credit/liquidity premium, especially for investment-grade bonds.
 - The CAPM looks like it fails for corporate bonds largely because it cannot price the duration component embedded in those standard excess returns.
 - Once duration is stripped out, corporate bond returns look much more compatible with simple equity-based pricing models.
- **Main decomposition result.** From 1986 to 2020, only about 9% of investment-grade standard excess returns and 37% of high-yield standard excess returns reflect credit/liquidity risk; the rest comes from the long-term Treasury component.
- **Main premium result.** The average duration-adjusted excess return is only 0.38% annualized for investment-grade bonds and 1.94% annualized for high-yield bonds, and neither estimate is statistically different from zero.
- **Main CAPM result.** The standard CAPM performs poorly on unadjusted corporate bond excess returns, especially for investment-grade portfolios. But once the dependent variable is the duration-adjusted return, the alphas become statistically insignificant and the model’s fit improves materially.
- **Main two-factor result.** A duration-adjusted CAPM that splits the equity market return into a Treasury-duration component and a dividend-risk component sharply improves the time-series fit of corporate bond returns.
- **Main comparison result.** For standard excess returns, the duration-adjusted bond CAPM (built from the decomposed corporate bond market return) produces extremely high time-series R^2 values and competes closely with more elaborate bond-factor models.

- **Main anomaly result.** Duration adjustment helps explain credit-spread, downside-risk, and illiquidity portfolios, but it does *not* explain the corporate bond momentum effect.
- **Economic takeaway.** In a nonstationary interest-rate environment, standard excess returns are a misleading object for corporate bond asset pricing. Much of what looks like compensation for risky bond investing is actually the return on long-duration safe assets.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- Corporate bonds are especially useful because their cash flows are much more explicitly defined than those of equities.
- That makes it possible to construct a much cleaner duration-matched government-bond counterfactual than in the stock market.
- The sample period is also ideal for the paper’s question: 1986–2020 is a long era of largely declining long-term nominal yields.

Interpretation. The paper’s argument is not merely that bond returns were high. It is that the *source* of those realized returns matters, and corporate bonds are a clean laboratory for separating interest-rate duration from corporate risk.

2.2 Data source and sample construction

The bond return sample is built from two quote databases:

- **January 1986 to June 1997:** Lehman/Warga Fixed Income database, using actively quoted bonds and dropping matrix prices.
- **July 1997 to December 2020:** Bank of America Merrill Lynch quote data, using the last quote of each month.

Sample restrictions are standard and important:

- U.S.-dollar denominated bonds;
- senior secured or senior unsecured;
- nonconvertible;
- fixed semiannual cash coupons;
- maturity between 1 and 30 years;
- exclude financials, utilities, and quasi-public issuers to keep the sample comparable across databases.

Additional details:

- the combined panel has 419 months;

- there is no valid bond price in April 1988, so the sample is one month short of a full 35 years;
- the sample tracks Bloomberg-Barclays investment-grade and high-yield indices extremely closely;
- the main results are also robust in TRACE data.

Interpretation. The authors are not working with a quirky corner of the bond market. The sample is broad, large, and very close to the investable corporate bond universe.

2.3 Portfolio construction and key sorts

The main test assets are value-weighted corporate bond portfolios sorted by credit rating:

- AA,
- A,
- BBB,
- BB,
- B,

with investment-grade (IG) and high-yield (HY) aggregates built from the rating spectrum.

The paper later extends the analysis to portfolios sorted by:

- credit spread,
- downside risk,
- illiquidity,
- issuer equity momentum.

Interpretation. The rating sorts are the natural starting point because they line up with both duration and credit risk. The later characteristic sorts test whether the paper’s message generalizes beyond ratings.

2.4 Main return definitions

The realized return on bond i is

$$r_{i,t} = \frac{P_{i,t} + AI_{i,t} + C_{i,t}}{P_{i,t-1} + AI_{i,t-1}} - 1,$$

where $P_{i,t}$ is the clean price, $AI_{i,t}$ accrued interest, and $C_{i,t}$ the coupon payment if a coupon is paid in month t .

The decomposition starts from Macaulay duration:

$$\text{Dur}_{i,t} = \sum_{k=1}^{\infty} w_{i,t,k} t_k,$$

where $w_{i,t,k}$ is the present-value weight of cash flow k and t_k is its time to payment.

For a coupon bond, the paper computes those weights from the full present-value schedule of coupons and principal, and then forms the duration-matched Treasury return:

$$r_{i,t+1}^{Tsy} = \sum_{k=1}^K w_{i,t,k} r_{t+1,k}^b,$$

where $r_{t+1,k}^b$ is the return on a zero-coupon Treasury strip at maturity k .

The key object is then the duration-adjusted return:

$$r_{i,t+1}^{DurAdj} = r_{i,t+1} - r_{i,t+1}^{Tsy}.$$

Interpretation. This is the core accounting move of the paper. Instead of asking “How much did the bond beat cash?”, the paper asks “How much did the bond beat a government bond portfolio with the same duration exposure?”

2.5 Equity-index decomposition

To build an equity-based two-factor model, the paper also decomposes the S&P 500 into a duration-driven Treasury component and a residual dividend-risk component.

Under a Gordon-growth structure,

$$S_t = \frac{D_t}{\mu - g},$$

so the index duration is

$$\text{Dur}_t = \frac{1}{\mu - g},$$

that is, the inverse of the dividend yield.

The paper converts this continuous-time representation into monthly weights for dividend strips and constructs a duration-matched Treasury return for the stock market,

$$r_{t+1}^{Tsy} = \sum_{k=1}^{\infty} w_{t,k} r_{t+1,k}^b,$$

and then defines the stock market’s duration-adjusted component as

$$r_{t+1}^{DurAdj} = r_{t+1}^{SP500} - r_{t+1}^{Tsy}.$$

Interpretation. This lets the authors ask whether corporate bonds are better explained by:

- a pure equity market factor;
- or by an equity market factor that has been split into “duration news” and “cash-flow/dividend-risk news.”

2.6 Other data inputs

The analysis also requires:

- factor returns from Amit Goyal’s and Ken French’s datasets;
- corrected Bai, Bali, and Wen (2019) bond factors from Dickerson, Mueller, and Robotti;
- zero-coupon Treasury yields from the Federal Reserve term-structure series based on Gürkaynak, Sack, and Wright (2007).

Interpretation. The paper does not only redefine test-asset returns. It also rethinks the factor side of the asset-pricing regressions.

2.7 Descriptive facts

The descriptive facts already point toward the paper’s main message.

From Table 1:

- investment-grade bonds have average yield 6.00%, average spread 1.36%, and duration 6.80;
- high-yield bonds have average yield 9.66%, average spread 5.16%, and duration 5.09;
- average monthly return is 0.61% for IG and 0.69% for HY;
- investment-grade realized returns exceeded promised yields over the sample, which is exactly what one expects in a falling-rate environment;
- higher-rated bonds generally have longer duration because they have longer maturities and lower coupons.

Interpretation. The surprising fact is not that high-yield spreads are larger. It is that *realized* returns are much more compressed across rating groups than promised yields suggest, which is already a hint that duration is doing a lot of the work.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper’s mechanism is conceptually simple:

- Standard excess returns over Treasury bills combine interest-rate duration with credit/liquidity risk.
- Over a period of persistent declines in long-term rates, long-duration assets receive large capital gains.
- Investment-grade corporate bonds have substantial duration exposure, so they inherit much of that windfall.
- If researchers treat those realized gains as “credit premia,” they will overstate the reward to bearing corporate risk and mis-evaluate factor models.

This leads to three main empirical predictions:

- duration-matched Treasury returns should explain a large share of corporate bond returns, especially for safer bonds;
- measured credit/liquidity premia should shrink sharply after duration adjustment;
- the CAPM's failure should be concentrated in the duration component rather than in the corporate-risk component.

3.2 Baseline CAPM

The paper first estimates the usual CAPM on standard excess returns:

$$r_{i,t} - r_{f,t} = \alpha + \beta(r_t^{SP500} - r_{f,t}) + \varepsilon_{i,t}.$$

This regression is run on:

- standard corporate bond excess returns;
- duration-matched Treasury returns of those same portfolios;
- duration-adjusted returns.

Interpretation. This is the cleanest test of the paper's claim. If the CAPM fails only on the unadjusted object but works on the duration-adjusted object, then the problem is not necessarily that equity factors cannot price corporate risk; it may simply be that they cannot price the long-term Treasury windfall embedded in standard excess returns.

3.3 Duration-adjusted CAPM

The paper's main alternative model is

$$r_{i,t} - r_{f,t} = \alpha + \beta_{Tsy}(r_t^{Tsy} - r_{f,t}) + \beta_{DurAdj}(r_t^{SP500} - r_t^{Tsy}) + \varepsilon_{i,t}.$$

The two factors are:

- a duration-matched Treasury component of the stock market return;
- a residual dividend-risk component of the stock market return.

Interpretation. The usual market factor is being decomposed into a bond-like duration factor and an equity-cash-flow factor. This is exactly what the paper argues is necessary when one studies risky assets over a long falling-rate period.

3.4 Duration-adjusted bond CAPM

To examine whether there is still value in using information specific to the corporate bond market, the paper also constructs a two-factor *duration-adjusted bond CAPM*. Conceptually, it replaces the decomposed S&P 500 factor with the decomposed value-weighted corporate bond market return:

$$r_{i,t} - r_{f,t} = \alpha + \beta_{Tsy}(r_t^{Tsy,bond} - r_{f,t}) + \beta_{DurAdj}r_t^{DurAdj,bond} + \varepsilon_{i,t}.$$

Interpretation. This preserves the paper’s duration logic but allows the factor to use bond-market information directly. It is useful for asking whether the remaining pricing structure is integrated with equities or still segmented enough that a bond-based factor works better.

3.5 Comparison benchmark: Bai, Bali, and Wen style model

The paper compares its duration-based models to the widely used bond-factor specification associated with Bai, Bali, and Wen:

- corporate bond market factor,
- downside risk factor,
- credit risk factor,
- liquidity risk factor.

Interpretation. The point is not to argue that these factors are useless. The point is that part of their apparent success on standard excess returns may just be coming from their ability to load on the Treasury-duration component.

3.6 Subperiod design

A crucial part of the paper is the split between:

- 1986–2000, when stock and Treasury returns are positively correlated;
- 2001–2020, when that correlation turns negative.

Interpretation. The CAPM’s apparent performance should depend heavily on whether Treasury returns move with or against the stock market. If the CAPM’s “failure” is really a duration problem, it should be much worse in the later negative-correlation era.

3.7 Additional specifications

The paper also studies:

- buy-and-hold portfolios sorted by initial rating rather than continuously resorted portfolios;
- splits by bond exit path (matured, called, defaulted, censored);
- splits by embedded optionality (noncallable, fixed call schedule, make-whole call);
- alternative duration-matching schemes;
- alternative portfolio sorts by spread, downside risk, illiquidity, and momentum.

Interpretation. These specifications are designed to show that the paper’s main message is not an artifact of one specific portfolio definition or one specific weighting scheme.

3.8 Estimation details

The time-series regressions use:

- monthly data;
- mostly 419 observations from January 1986 to December 2020;
- Newey-West standard errors with 12 lags;
- Gibbons-Ross-Shanken tests to assess whether alphas are jointly zero.

Interpretation. This is standard asset-pricing econometrics. The novelty is not the estimator but the redefinition of returns and factors.

4 Identification and what makes the estimates credible

4.1 What identifies the main claims

The central object is not a treatment effect from an external shock. It is an accounting-and-asset-pricing decomposition.

Credibility comes from three ingredients:

- observed bond cash flows, prices, and maturities;
- a duration-based Treasury benchmark built from the term structure;
- comparison of asset-pricing model performance before and after removing the duration component.

Interpretation. The paper is not saying “interest rates causally changed firm behavior.” It is saying “you are measuring the wrong excess return object if you ignore duration.”

4.2 Why duration contamination is the central confound

The whole paper is motivated by a particular historical fact:

- from the mid-1980s to 2020, long-term Treasury yields fell dramatically;
- long-duration assets therefore earned large ex post gains;
- corporate bonds, especially investment-grade bonds, carry a lot of duration.

If one subtracts only the Treasury-bill rate, then the remaining excess return includes both:

- the long-term safe-rate windfall;
- and the corporate-risk premium.

Interpretation. This is exactly the wrong benchmark when the research question is about compensation for bearing corporate credit and liquidity risk.

4.3 Why corporate bonds are a strong laboratory

The paper's decomposition is especially persuasive in corporate bonds because:

- cash flows are much more explicit than in equities;
- duration can therefore be measured more precisely;
- rating portfolios naturally line up with both maturity structure and credit risk.

Interpretation. If the same duration logic matters in equities, it should matter at least as much here, where the fixed-income cash-flow structure is cleaner.

4.4 Alternative weighting schemes

The paper checks two alternative ways of matching Treasury returns:

- maturity matching instead of duration matching;
- regression-based sensitivity weights.

The paper's interpretation of those alternatives is revealing:

- maturity matching tends to *overstate* the interest-rate component because it overweights distant cash flows;
- regression-based matching tends to *understate* the interest-rate component because negative co-movement between Treasury yields and credit spreads attenuates the measured Treasury beta.

Interpretation. The preferred duration-based construction sits in the economically sensible middle ground.

4.5 Embedded options, defaults, and censoring

A key issue is that average credit spreads are much larger than realized duration-adjusted excess returns. The paper argues that this wedge comes from:

- default losses;
- issuer callability / prepayment when rates fall;
- censoring because some bonds remain outstanding at the end of the sample.

Interpretation. This is important because it explains why a bond can trade at a sizable spread yet deliver a low realized premium ex post. The spread is not a guaranteed premium; it is partly compensation for option exercise and default events that materialize unfavorably for investors.

4.6 The stock-bond correlation regime change

The subperiod split is not just a robustness check. It is a direct diagnostic.

The paper shows:

- when stock and Treasury returns are positively correlated (1986–2000), standard CAPM performance on unadjusted bond returns looks less bad;
- when the correlation turns negative (2001–2020), the CAPM’s failure on unadjusted investment-grade bond returns becomes severe;
- on duration-adjusted returns, the opposite pattern emerges: the CAPM works especially well in the later period.

Interpretation. This is strong evidence that the CAPM “failure” is about bond-duration contamination, not about an inability to price corporate risk per se.

4.7 Market integration versus segmentation

The paper reads its results carefully:

- the success of the equity-based duration-adjusted CAPM supports a meaningful degree of integration between corporate bond risk and equity risk;
- but the even better time-series fit of the bond-based duration-adjusted CAPM leaves room for segmentation or bond-specific information.

Interpretation. The paper does not claim complete market integration. It claims that once duration is handled properly, the gap between equity-based and bond-based models narrows sharply.

4.8 Main limitations

- The duration calculation ignores embedded options in the bond indenture, so the paper acknowledges that it likely overstates the role of interest-rate duration somewhat.
- The realized-return evidence comes from a historically unusual sample with a long secular fall in yields.
- The paper shows that duration adjustment matters greatly in this sample, but it does not claim the exact same magnitudes will hold in a rising-rate era.
- The paper’s strongest results are about *measuring* risk premia and *evaluating* factor models, not about deriving a fully structural model of bond pricing.

5 Tables and figures

5.1 Figure 1: the secular fall in zero-coupon Treasury yields

Read:

- The 1-year, 10-year, and 30-year zero-coupon Treasury yields all trend downward over the sample.
- The long end of the curve falls especially dramatically.

- The figure visually explains why long-duration assets enjoyed large ex post valuation gains.

Economic message. Figure 1 is the paper’s motivating macro fact. Without the long decline in risk-free discount rates, standard excess returns would not be nearly as contaminated by duration windfalls.

5.2 Table 1: summary statistics and why safer bonds have more duration

Read:

- Investment-grade bonds have average yield 6.00%, spread 1.36%, and duration 6.80.
- High-yield bonds have average yield 9.66%, spread 5.16%, and duration 5.09.
- Monthly total returns are surprisingly similar across ratings: 0.61% for IG and 0.69% for HY.
- In the benchmark-factor panel, the S&P 500’s duration-matched Treasury component has a mean monthly return of 1.06%, while its duration-adjusted component is slightly negative on average.

Economic message. Safer bonds have more duration. That means a falling-rate sample will mechanically flatter their realized returns even if the corporate-risk premium itself is small.

5.3 Table 2 and Figure 2: the decomposition of corporate bond returns

Read:

- For investment-grade bonds, the mean monthly total return is 0.608%, the Treasury-bill rate is 0.253%, and the standard excess return is 0.355%.
- But the duration-matched Treasury return is 0.576%, leaving only 0.032% per month of duration-adjusted excess return.
- For high-yield bonds, the standard excess return is 0.439% per month, but the duration-adjusted excess return is only 0.162%.
- Figure 2 shows that, for investment-grade bonds, cumulative total returns barely exceed cumulative duration-matched Treasury returns; the residual duration-adjusted series stays close to zero for most of the sample.
- For high-yield bonds, the duration-adjusted component matters more, but even there the Treasury component still explains a large share of total performance.

Economic message. Table 2 and Figure 2 are the empirical core of the paper. Standard excess returns mostly capture long-duration government-bond gains, not corporate credit premia.

Table 1
Summary statistics on corporate bond portfolios and benchmark returns

A. Credit rating portfolio characteristics						
Rating	Yield	Spread	Duration	Fixed call	Make-Whole	Bond count
AA	5.43	0.83	6.74	0.34	0.28	150
A	5.74	1.10	6.90	0.34	0.37	614
BBB	6.41	1.76	6.67	0.40	0.42	879
BB	7.83	3.31	5.42	0.62	0.22	348
B	10.10	5.61	4.91	0.92	0.04	428
IG	6.00	1.36	6.80	0.28	0.36	1,677
HY	9.66	5.16	5.09	0.72	0.13	932

B. Monthly returns of credit rating portfolios								
	Mean	SD	Min.	p10	p50	p90	Max.	Obs.
AA	0.60	1.49	-5.55	-1.27	0.64	2.33	7.51	419
A	0.60	1.54	-8.00	-1.22	0.62	2.40	7.47	419
BBB	0.62	1.69	-10.65	-1.07	0.64	2.49	6.71	419
BB	0.70	1.93	-14.90	-1.45	0.83	2.58	7.72	419
B	0.66	2.42	-15.92	-1.69	0.86	2.78	11.64	419
IG	0.61	1.59	-8.55	-1.18	0.58	2.42	6.26	419
HY	0.69	2.27	-16.72	-1.51	0.84	2.72	11.06	419

C. Monthly benchmark returns								
	Mean	SD	Min.	p10	p50	p90	Max.	Obs.
S&P 500	0.98	4.39	-21.58	-4.54	1.36	6.03	13.52	419
S&P 500 Treasury	1.06	5.35	-15.01	-5.25	0.99	7.24	28.61	419
S&P 500 dur-adj.	-0.07	7.20	-38.81	-9.39	0.54	7.89	17.75	419
Treasury bill	0.25	0.21	0.00	0.00	0.24	0.51	0.79	419
Corp. bond market	0.47	1.87	-9.29	-1.01	0.43	2.26	10.81	197
Corp. bond DRF	0.65	3.24	-15.90	-2.18	0.56	3.07	16.77	197
Corp. bond CRF	0.38	3.20	-21.91	-2.42	0.31	3.45	13.23	197
Corp. bond LRF	0.34	1.37	-5.08	-0.78	0.25	1.61	8.72	197

This table reports summary statistics on the corporate bond portfolio and factor benchmark returns. Except for the factors of [Bai, Bali, and Wen \(2019\)](#), sample consists of 419 monthly observations from January 1986 to December 2020. Panel A presents the average characteristics of corporate bond portfolios sorted by credit rating. Yield and Spread, the difference between yield-to-maturity (without option adjustment) and the duration-matched Treasury yield, are value-weighted averages in percentage terms. Duration is also a value-weighted average. Fixed call and Make-Whole are the fraction of bonds with fixed-price and make-whole call options, respectively. Bond count is the average number of bonds in each rating-month. The bottom two rows group investment-grade (IG) and high-yield (HY) bonds, including the relatively small number of bonds in the AAA and CCC categories that are omitted from the individual rating portfolios. Panel B summarizes the value-weighted returns of each portfolio, in percentage terms. Panel C summarizes the returns of the benchmark portfolios used in our analysis. S&P 500 Treasury and duration-adjusted returns are based on the decomposition in [Van Binsbergen \(2021\)](#). Corp. bond market, DRF (downside risk), CRF (credit risk), and LRF (liquidity risk) are the risk factors from [Bai, Bali, and Wen \(2019\)](#) and start from August 2004 (197 monthly observations).

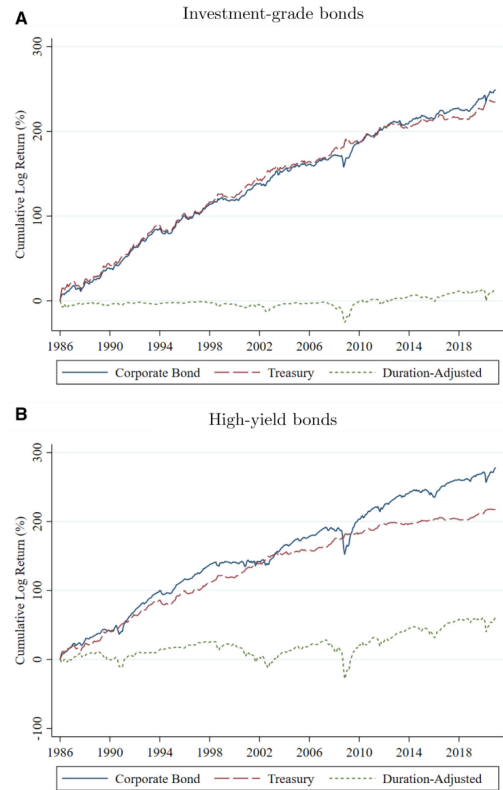


Figure 2
Cumulative log returns of corporate bond portfolios
This figure presents the cumulative log returns of corporate bond portfolios sorted by credit rating. Bonds are sorted into investment-grade (BBB- and higher) and high-yield (BB+ and lower) value-weighted portfolios. Corporate Bond is the realized portfolio return. Treasury is the duration-matched Treasury return defined in Section 1.1. Duration-Adjusted is the difference between the corporate bond return and the duration-matched Treasury return.

Table 2
Decomposition of corporate bond returns

A. Return decomposition						
Rating	Return $r_{i,t}$	Standard approach		Duration-adjusted approach		
		Treasury bill $r_{f,t}$	Excess $r_{i,t} - r_{f,t}$	Treasury $r_{i,t}^{Tsy}$	Excess $r_{i,t}^{DurAdj}$	$\log(1 + r_{i,t}^{DurAdj})$
AA	0.598	0.253	0.344	0.573	0.025	0.020
A	0.598	0.253	0.345	0.577	0.020	0.014
BBB	0.622	0.253	0.369	0.572	0.050	0.038
BB	0.703	0.253	0.450	0.540	0.164	0.140
B	0.660	0.253	0.407	0.525	0.135	0.095
IG	0.608	0.253	0.355	0.576	0.032	0.023
HY	0.692	0.253	0.439	0.530	0.162	0.126
B. Buy-and-hold strategy						
Rating	Return $r_{i,t}$	Standard approach		Duration-adjusted approach		
		Treasury bill $r_{f,t}$	Excess $r_{i,t} - r_{f,t}$	Treasury $r_{i,t}^{Tsy}$	Excess $r_{i,t}^{DurAdj}$	$\log(1 + r_{i,t}^{DurAdj})$
AA	0.600	0.253	0.347	0.572	0.029	0.024
A	0.610	0.253	0.357	0.575	0.035	0.028
BBB	0.648	0.253	0.395	0.572	0.076	0.063
BB	0.689	0.253	0.436	0.538	0.151	0.130
B	0.654	0.253	0.400	0.522	0.132	0.093
IG	0.622	0.253	0.369	0.576	0.047	0.038
HY	0.649	0.253	0.396	0.524	0.125	0.094
C. Sample split by exit path						
Rating	Return $r_{i,t}$	Standard approach		Duration-adjusted approach		
		Treasury bill $r_{f,t}$	Excess $r_{i,t} - r_{f,t}$	Treasury $r_{i,t}^{Tsy}$	Excess $r_{i,t}^{DurAdj}$	Sample share
<i>Matured</i>						
IG	0.527	0.253	0.273	0.477	0.050	0.440
HY	0.752	0.253	0.498	0.466	0.285	0.244
<i>Called early</i>						
IG	0.593	0.253	0.340	0.590	0.003	0.059
HY	0.868	0.253	0.615	0.531	0.336	0.138
<i>Defaulted</i>						
IG	0.167	0.253	-0.086	0.620	-0.453	0.010
HY	-0.751	0.253	-1.004	0.546	-1.298	0.063
<i>Censored</i>						
IG	0.680	0.253	0.427	0.654	0.026	0.491
HY	0.746	0.253	0.492	0.551	0.195	0.554

Table 2
Continued

D. Sample split by embedded optionality						
Rating	Return $r_{i,t}$	Standard approach		Duration-adjusted approach		Sample share
		Treasury bill $r_{f,t}$	Excess $r_{i,t} - r_{f,t}$	Treasury $r_{i,t}^{Tsy}$	Excess $r_{i,t}^{DurAdj}$	
<i>Noncallable</i>						
IG	0.611	0.253	0.357	0.545	0.066	0.366
HY	0.762	0.253	0.509	0.513	0.250	0.150
<i>Fixed call schedule</i>						
IG	0.607	0.253	0.353	0.629	-0.022	0.279
HY	0.664	0.253	0.411	0.531	0.133	0.723
<i>Make-whole call</i>						
IG	0.511	0.253	0.258	0.492	0.019	0.355
HY	0.811	0.253	0.558	0.476	0.335	0.127

This table summarizes the decomposed returns of corporate bond portfolios sorted by credit rating. The sample consists of 419 monthly observations from January 1986 to December 2020. Panel A is based on portfolios rebalanced each month; panel B is based on portfolios sorted by the bond's initial rating and not rebalanced; panel C splits the sample by exit path, that is, matured, called, defaulted, or censored (still outstanding at the end of 2020); and panel D splits the sample by the presence of embedded options, that is, noncallable bonds, bonds with a fixed-price call schedule, and bonds without a fixed-price call but with a make-whole call provision. From left to right, the columns contain the average monthly return for each corporate bond portfolio; the 1-month Treasury-bill return and the standard excess corporate bond return; the duration-matched Treasury return and the duration-adjusted excess corporate bond return; and the log duration-adjusted excess corporate bond return (to account for compounding). In panels C and D, the rightmost column reports the average fraction of bonds in each subsample.

5.4 Table 2, panels C and D: why spreads exceed realized duration-adjusted returns

Read:

- Bonds that matured delivered higher duration-adjusted returns than called, defaulted, or censored bonds.
- For investment-grade bonds, called bonds earned essentially zero duration-adjusted excess return.
- Noncallable bonds outperform bonds with fixed call schedules.
- The wedge between spreads and realized returns is especially associated with falling rates and issuer prepayment.

Economic message. The paper is not claiming spreads are “wrong.” It is showing that ex post realized premia are smaller because issuers call debt when it is in their interest to refinance and because some bonds default or remain outstanding at the sample end.

5.5 Table 3: why the standard CAPM appears to fail

Read:

- For standard excess returns, investment-grade CAPM alphas are economically large: 0.305% per month for AA, 0.290% for A, and 0.261% for BBB.

- The adjusted R^2 is tiny for safer bonds: 2.3% for AA and 4.3% for A.
- For the duration-matched Treasury component, CAPM alphas remain sizable and R^2 is approximately zero.
- For duration-adjusted returns, the alphas become small and statistically insignificant, and the GRS test no longer rejects.

Economic message. The CAPM’s “failure” is living in the Treasury-duration part of bond returns. Once that piece is removed, the remaining corporate-risk return is much less anomalous.

Table 3
CAPM regressions of corporate bond returns

A. Corporate bond returns					
Rating	AA	A	BBB	BB	B
β	0.054 (2.11)	0.074 (2.61)	0.148 (4.28)	0.253 (5.96)	0.326 (6.84)
α	0.305 (3.89)	0.290 (3.66)	0.261 (3.03)	0.265 (2.95)	0.169 (1.47)
Adj. R^2	.023	.043	.146	.328	.345
Mean $ \alpha $	0.258				
GRS p -value	.0003				

B. Duration-matched Treasury returns					
Rating	AA	A	BBB	BB	B
β	-0.015 (-0.43)	-0.021 (-0.61)	-0.023 (-0.72)	-0.010 (-0.36)	-0.010 (-0.36)
α	0.331 (3.86)	0.339 (3.98)	0.336 (4.05)	0.294 (3.97)	0.279 (4.01)
Adj. R^2	-.001	.000	.001	-.001	-.001
Mean $ \alpha $	0.316				
GRS p -value	.001				

C. Duration-adjusted returns					
Rating	AA	A	BBB	BB	B
β	0.069 (3.12)	0.095 (3.37)	0.171 (3.99)	0.263 (5.02)	0.336 (5.65)
α	-0.025 (-0.60)	-0.049 (-0.97)	-0.075 (-0.97)	-0.029 (-0.31)	-0.110 (-0.89)
Adj. R^2	.107	.148	.228	.291	.271
Mean $ \alpha $	0.058				
GRS p -value	.360				

This table presents regressions of the excess returns of corporate bond portfolios sorted by credit rating on the excess return of the S&P 500 Index:

$$r_{i,t} - r_{f,t} = \alpha + \beta(r_{SP500,t} - r_{f,t}) + e_{i,t}.$$

The sample consists of 419 monthly observations from January 1986 to December 2020. Value-weighted returns by credit rating are computed using bond-level quote data from Lehman Brothers and Bank of America Merrill Lynch. Panel A uses excess corporate bond returns as the dependent variable; panel B uses the duration-matched Treasury excess return from Equation (4); and panel C uses the duration-adjusted return from Equation (6). Excess returns are computed by subtracting the 1-month Treasury-bill rate. t -statistics based on Newey and West (1987) standard errors with 12 lags are reported in parentheses. GRS p -value is from the Gibbons, Ross, and Shanken (1989) test.

Table 4
Subperiod analysis: The role of equity Treasury correlation

A. Corporate bond returns A1. 1986–2000					
Rating	AA	A	BBB	BB	B
β	0.118 (2.84)	0.116 (2.90)	0.125 (3.70)	0.149 (4.27)	0.211 (4.83)
α	0.169 (1.41)	0.176 (1.53)	0.171 (1.48)	0.228 (1.95)	0.145 (0.78)
Adj. R^2	.111	.117	.156	.199	.223
Mean $ \alpha $	0.178				
GRS p -value	.316				

A2. 2001–2020					
Rating	AA	A	BBB	BB	B
β	0.005 (0.14)	0.043 (0.96)	0.167 (3.23)	0.336 (7.06)	0.418 (7.35)
α	0.392 (4.33)	0.365 (3.62)	0.333 (2.83)	0.315 (2.52)	0.211 (1.47)
Adj. R^2	-.004	.010	.146	.430	.440
Mean $ \alpha $	0.323				
GRS p -value	.0001				

B. Duration-adjusted returns B1. 1986–2000					
Rating	AA	A	BBB	BB	B
β	-0.010 (-0.38)	-0.004 (-0.16)	0.012 (0.42)	0.043 (1.50)	0.111 (1.85)
α	-0.057 (-1.25)	-0.051 (-1.00)	-0.051 (-0.80)	0.009 (0.10)	-0.065 (-0.35)
Adj. R^2	-.003	-.005	-.002	.020	.043
Mean $ \alpha $	0.047				
GRS p -value	.719				

B2. 2001–2020					
Rating	AA	A	BBB	BB	B
β	0.131 (6.01)	0.173 (5.99)	0.295 (6.77)	0.436 (10.48)	0.513 (9.94)
α	0.014 (0.23)	-0.027 (-0.37)	-0.058 (-0.54)	-0.010 (-0.08)	-0.096 (-0.66)
Adj. R^2	.321	.361	.449	.521	.488
Mean $ \alpha $	0.040				
GRS p -value	.340				

This table presents regressions of the excess returns of corporate bond portfolios sorted by credit rating on the excess return of the S&P 500 Index:

$$r_{i,t} - r_{f,t} = \alpha + \beta(r_{SP500,t} - r_{f,t}) + e_{i,t}.$$

The sample consists of 419 monthly observations from January 1986 to December 2020. Subpanels of each panel consider the subperiods 1986–2000 and 2001–2020, respectively, to assess how the change in the correlation between equity and Treasury returns from positive to negative (Campbell, Pflueger, and Viceira 2020) affects the asset pricing results. Value-weighted returns by credit rating are computed using bond-level quote data from Lehman Brothers and Bank of America Merrill Lynch. Panel A uses excess corporate bond returns as the dependent variable, while panel B uses the duration-adjusted return from Equation (6). Excess returns are computed by subtracting the 1-month Treasury-bill rate. t -statistics based on Newey and West (1987) standard errors with 12 lags are reported in parentheses. GRS p -value is from the Gibbons, Ross, and Shanken (1989) test.

5.6 Table 4: the stock-bond correlation regime matters

Read:

- In 1986–2000, the CAPM does a tolerable job on standard bond excess returns because stocks and Treasuries move together.
- In 2001–2020, CAPM fit for unadjusted investment-grade returns collapses: the alphas become highly significant and the R^2 for AA and A is near zero.
- On duration-adjusted returns, the later period is exactly when the CAPM works better.

Economic message. This table shows that the result is not a generic statement that “the CAPM does not price bonds.” It is a statement about how duration interacts with the sign of the stock-Treasury correlation.

5.7 Table 5 and Table 6: the duration-adjusted CAPM

Table 5
Duration-adjusted CAPM regressions of corporate bond returns

A. Corporate bond returns					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	0.291 (0.81)	0.304 (0.80)	0.328 (0.77)	0.328 (0.79)	0.325 (0.82)
$\hat{\rho}_{DurAdj}$	0.075 (0.76)	0.095 (0.75)	0.164 (0.55)	0.260 (0.30)	0.326 (0.05)
α	0.116 (1.99)	0.107 (1.60)	0.118 (1.45)	0.205 (2.30)	0.170 (1.46)
Adj. R^2	.626	.565	.410	.362	.343
CAPM adj. R^2	.023	.043	.146	.328	.345
Incremental R^2	.603	.522	.264	.034	-.002
Mean $ \alpha $	0.143				
GRS p -value	.029				

B. Duration-matched Treasury returns					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	0.318 (0.86)	0.313 (2.02)	0.291 (18.89)	0.248 (14.42)	0.225 (12.77)
$\hat{\rho}_{DurAdj}$	0.035 (1.36)	0.010 (0.80)	0.006 (0.51)	0.014 (1.08)	0.012 (0.87)
α	0.066 (1.69)	0.073 (1.92)	0.086 (2.21)	0.088 (1.99)	0.092 (2.08)
Adj. R^2	.845	.852	.801	.748	.693
CAPM adj. R^2	-.001	-.000	.001	-.001	-.001
Incremental R^2	.846	.852	.800	.749	.694
Mean $ \alpha $	0.081				
GRS p -value	.084				

C. Duration-adjusted returns					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	-0.026 (-0.06)	-0.009 (-0.30)	0.037 (0.01)	0.080 (2.09)	0.100 (2.09)
$\hat{\rho}_{DurAdj}$	0.000 (3.47)	0.005 (0.60)	0.158 (4.32)	0.247 (5.59)	0.314 (6.26)
α	0.080 (1.34)	0.034 (0.76)	0.032 (0.47)	0.117 (1.50)	0.078 (0.69)
Adj. R^2	.357	.365	.399	.461	.434
CAPM adj. R^2	.307	.148	.228	.291	.271
Incremental R^2	.250	.217	.171	.170	.163
Mean $ \alpha $	0.062				
GRS p -value	.121				

This table presents time-series regressions of the excess returns of corporate bond portfolios sorted by credit rating on the duration-matched Treasury and dividend risk components of the S&P 500 Index excess return:

$$r_{i,t} - r_{f,t} = \alpha + \hat{\rho}_{T,t} (r_{T,t} - r_{f,t}) + \hat{\rho}_{DurAdj,t} (r_{DurAdj,t} - r_{f,t}) + \hat{\rho}_{DivRisk,t} (r_{DivRisk,t} - r_{f,t}) + \epsilon_{i,t}$$

The sample consists of 419 monthly observations from January 1986 to December 2020. Value-weighted returns by credit rating are computed using bond-level quote data from Lehman Brothers and Bank of America Merrill Lynch. Panel A uses excess corporate bond returns as the dependent variable; panel B uses the duration-matched Treasury excess return from Equation (4); and panel C uses the duration-adjusted return from Equation (5). Excess returns are computed by subtracting the 1-month Treasury bill rate. t -statistics based on Newey and West (1987) standard errors with 12 lags are reported in parentheses. GRS p -value is from the Gibbons, Ross, and Shanken (1989) test.

Table 6
Subperiod analysis of the duration-adjusted CAPM

A. Corporate bond returns A1: 1986-2000					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	0.276 (0.43)	0.264 (0.41)	0.248 (0.71)	0.252 (0.70)	0.241 (0.70)
$\hat{\rho}_{DurAdj}$	0.055 (3.56)	0.057 (3.54)	0.076 (4.74)	0.108 (4.80)	0.109 (4.52)
α	0.034 (0.50)	0.050 (0.73)	0.066 (0.81)	0.140 (1.35)	0.119 (0.65)
Adj. R^2	.733	.711	.617	.487	.232
CAPM adj. R^2	.111	.117	.156	.199	.223
Incremental R^2	.622	.594	.461	.288	.009
Mean $ \alpha $	0.082				
GRS p -value	.555				

A2: 2001-2020					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	0.313 (7.20)	0.363 (8.19)	0.438 (7.65)	0.411 (7.61)	0.429 (8.00)
$\hat{\rho}_{DurAdj}$	0.003 (2.84)	0.135 (1.61)	0.245 (3.27)	0.358 (8.20)	0.421 (8.52)
α	0.174 (2.00)	0.160 (1.37)	0.142 (1.08)	0.282 (1.86)	0.203 (1.30)
Adj. R^2	.542	.492	.391	.441	.438
CAPM adj. R^2	-.004	.010	.146	.430	.440
Incremental R^2	.546	.482	.245	.011	-.002
Mean $ \alpha $	0.184				
GRS p -value	.005				

B. Duration-adjusted returns B1: 1986-2000					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	-0.081 (-2.86)	-0.079 (-2.89)	-0.068 (-2.73)	-0.044 (-2.21)	-0.032 (-1.66)
$\hat{\rho}_{DurAdj}$	0.018 (1.30)	0.026 (2.23)	0.044 (2.76)	0.078 (3.40)	0.168 (3.38)
α	0.004 (0.13)	0.013 (0.39)	0.017 (0.34)	0.084 (1.05)	0.058 (0.33)
Adj. R^2	.484	.524	.466	.324	.278
CAPM adj. R^2	-.003	-.005	-.002	.020	.043
Incremental R^2	.487	.529	.468	.304	.235
Mean $ \alpha $	0.035				
GRS p -value	.790				

Table 6
Continued

B2: 2001-2020					
Rating	AA	A	BBB	BB	B
$\hat{r}_{T,t}$	0.056 (2.06)	0.093 (2.58)	0.180 (3.02)	0.240 (4.18)	0.281 (4.54)
$\hat{\rho}_{DurAdj}$	0.109 (3.09)	0.150 (3.09)	0.262 (5.84)	0.380 (9.17)	0.446 (9.01)
α	0.067 (1.02)	0.030 (0.38)	0.024 (0.30)	0.129 (1.03)	0.068 (0.46)
Adj. R^2	.386	.408	.490	.586	.549
CAPM adj. R^2	.321	.361	.449	.521	.488
Incremental R^2	.065	.047	.041	.065	.061
Mean $ \alpha $	0.064				
GRS p -value	.084				

This table presents time-series regressions of the excess returns of corporate bond portfolios sorted by credit rating on the duration-matched Treasury and dividend risk components of the S&P 500 Index excess return:

$$r_{i,t} - r_{f,t} = \alpha + \hat{\rho}_{T,t} (r_{T,t} - r_{f,t}) + \hat{\rho}_{DurAdj,t} (r_{DurAdj,t} - r_{f,t}) + \hat{\rho}_{DivRisk,t} (r_{DivRisk,t} - r_{f,t}) + \epsilon_{i,t}$$

The sample consists of 419 monthly observations from January 1986 to December 2020. Subpanels of each panel consider the subperiods 1986-2000 and 2001-2020, respectively, to assess how the change in the correlation between equity and Treasury returns from positive to negative (Campbell, Philpott, and Vayns 2020) affects the asset pricing results. Value-weighted returns by credit rating are computed using bond-level quote data from Lehman Brothers and Bank of America Merrill Lynch. Panel A uses excess corporate bond returns as the dependent variable, while panel B uses the duration-adjusted return from Equation (5). Excess returns are computed by subtracting the 1-month Treasury bill rate. t -statistics based on Newey and West (1987) standard errors with 12 lags are reported in parentheses. GRS p -value is from the Gibbons, Ross, and Shanken (1989) test.

Read:

- For standard excess returns, the duration-adjusted CAPM raises adjusted R^2 for AA-rated bonds from 0.023 to 0.626 and for A-rated bonds from 0.043 to 0.565.
- Across the rating spectrum, the mean absolute alpha falls from 0.258% in the standard CAPM to 0.143%.
- For duration-adjusted returns, adjusted R^2 rises by roughly 16 to 25 percentage points relative to the plain CAPM.
- In the subperiod analysis, the same two-factor adjustment improves fit in both regimes, especially where the one-factor CAPM struggles.

Economic message. Once the market factor itself is decomposed into a duration piece and a dividend-risk piece, the equity-based model becomes much more plausible for corporate bonds.

5.8 Table 7: characteristic-sorted portfolios and the special case of momentum

Read:

- The duration-adjusted CAPM shrinks CAPM alphas for portfolios sorted by credit spread, downside risk, and illiquidity.
- For example, in the credit-spread sorts the CAPM mean absolute alpha of 0.254% falls to 0.144%.

Table 7
Duration-adjusted CAPM and other bond characteristics

A. Portfolios sorted by credit spread						
	Low	2	3	4	High	High - Low
Avg. $r_{i,t} - r_{f,t}$	0.293 (4.38)	0.360 (4.34)	0.387 (4.01)	0.458 (4.38)	0.494 (2.43)	0.201 (1.07)
α	0.083 (1.50)	0.101 (1.43)	0.117 (1.32)	0.201 (2.13)	0.218 (1.35)	0.135 (0.95)
Adj. R^2	.602	.547	.413	.354	.349	.433
Mean $ \alpha $	0.144					
CAPM α	0.256 (3.59)	0.293 (3.43)	0.273 (2.94)	0.267 (2.90)	0.183 (1.13)	-0.073 (-0.47)
CAPM adj. R^2	.024	.056	.140	.317	.347	.273
CAPM mean $ \alpha $	0.254					
B. Portfolios sorted by downside risk						
	Low	2	3	4	High	High - Low
Avg. $r_{i,t} - r_{f,t}$	0.229 (6.30)	0.323 (5.28)	0.376 (4.53)	0.458 (4.18)	0.637 (3.38)	0.407 (2.50)
α	0.151 (3.94)	0.162 (2.76)	0.134 (1.77)	0.129 (1.31)	0.260 (1.59)	0.109 (0.80)
Adj. R^2	.290	.400	.452	.477	.327	.304
Mean $ \alpha $	0.167					
CAPM α	0.202 (5.09)	0.266 (4.11)	0.287 (3.40)	0.326 (3.06)	0.378 (2.42)	0.176 (1.35)
CAPM adj. R^2	.063	.090	.110	.144	.271	.282
CAPM mean $ \alpha $	0.292					
C. Portfolios sorted by illiquidity						
	Low	2	3	4	High	High - Low
Avg. $r_{i,t} - r_{f,t}$	0.309 (4.14)	0.399 (3.89)	0.497 (3.97)	0.627 (3.97)	0.742 (3.36)	0.433 (2.73)
α	0.097 (1.31)	0.112 (1.10)	0.137 (1.13)	0.178 (1.23)	0.221 (1.05)	0.124 (0.79)
Adj. R^2	.393	.422	.433	.450	.427	.351
Mean $ \alpha $	0.149					
CAPM α	0.237 (3.14)	0.290 (2.90)	0.356 (3.01)	0.434 (3.05)	0.469 (2.57)	0.232 (1.89)
CAPM adj. R^2	.108	.147	.161	.203	.274	.295
CAPM mean $ \alpha $	0.357					

- Long-minus-short alpha spreads become statistically insignificant for those characteristic sorts.
- The major exception is issuer equity momentum: the duration-adjusted CAPM still leaves a highly significant high-minus-low alpha.

Economic message. The paper is not saying every bond anomaly is fake. It is saying several prominent ones are heavily contaminated by duration, while momentum looks more like a genuine underreaction-type anomaly.

5.9 Table 8 and Table 9: comparison with bond-factor models

Table 8
Model performance in the time series: Credit rating portfolios

A. Corporate bond returns												
CAPM			Dur.-adj. CAPM			Dur.-adj. bond CAPM			Bai et al. 2019 model			
	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2
AA	0.341	3.44	.011	0.042	0.03	.258	-0.002	-0.10	.897	0.108	2.46	.851
A	0.298	2.72	.040	0.039	0.36	.488	-0.066	-3.35	.953	0.011	0.30	.898
BBB	0.257	2.00	.211	0.054	0.35	.403	-0.038	-1.31	.961	-0.032	-0.56	.829
BB	0.178	1.41	.404	0.135	0.85	.497	0.124	3.72	.901	0.027	0.84	.875
B	0.092	0.58	.491	0.112	0.61	.490	0.106	2.57	.904	-0.042	-1.56	.928
Mean abs. value	0.233	2.03	.249	0.064	0.62	.487	0.067	2.21	.923	0.052	1.14	.876
GRS p -value	.008		.157			.011						
B. Duration-matched Treasury returns												
CAPM			Dur.-adj. CAPM			Dur.-adj. bond CAPM			Bai et al. 2019 model			
	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2
AA	0.380	3.63	.005	0.057	1.18	.844	-0.012	-0.91	.984	0.186	2.84	.681
A	0.394	3.64	.098	0.066	1.36	.845	-0.012	-1.59	.996	0.183	2.74	.677
BBB	0.391	3.68	.099	0.077	1.54	.828	-0.004	-0.89	.999	0.185	2.80	.674
BB	0.365	3.98	.091	0.094	1.85	.883	0.017	1.22	.969	0.157	2.67	.631
B	0.281	3.96	.091	0.098	1.89	.823	0.024	1.22	.935	0.150	2.60	.588
Mean abs. value	0.350	3.78	.095	0.078	1.56	.764	0.014	1.17	.977	0.172	2.73	.650
GRS p -value	.003		.074			.586			.048			
C. Duration-adjusted returns												
CAPM			Dur.-adj. CAPM			Dur.-adj. bond CAPM			Bai et al. 2019 model			
	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2	α	t -stat	Adj. R^2
AA	-0.040	-0.56	.376	0.025	0.51	.433	0.010	0.46	.857	-0.078	-2.50	.581
A	-0.096	-1.21	.388	-0.027	-0.20	.430	-0.054	-2.91	.939	-0.172	-3.82	.673
BBB	-0.134	-1.11	.407	-0.024	-0.17	.517	-0.034	-1.11	.964	-0.217	-3.23	.680
BB	-0.127	-0.97	.543	-0.041	-0.28	.609	0.107	3.41	.930	-0.189	-1.75	.821
B	-0.189	-1.10	.507	0.014	0.08	.574	0.082	1.89	.929	-0.232	-2.39	.874
Mean abs. value	0.117	0.99	.456	0.026	0.22	.513	0.057	1.96	.924	0.166	2.14	.726
GRS p -value	.485		.219			.011			.000			

This table presents time-series regressions of excess corporate bond returns on the CAPM, the duration-adjusted CAPM based on equity and bond market returns, and the pricing factors from Bai, Ball, and Wei (2019). The sample consists of 197 monthly observations from August 2004 to December 2020 only because of the availability of the Bai, Ball, and Wei (2019) factors. The dependent variables in panels A, B, and C are based on the excess returns of corporate bond portfolios, duration-matched Treasury returns, and duration-adjusted returns, respectively. Value-weighted index returns by credit rating are computed using bond level quote data from Lehman Brothers and Bank of America Merrill Lynch. Excess returns are computed by subtracting the 1-month Treasury-bill rate. t -statistics are based on Newey and West (1987) standard errors with 12 lags. GRS p -value is from the Gibbons, Ross, and Shanken (1989) test.

Read:

- In Table 8, the duration-adjusted bond CAPM achieves extremely strong time-series fit for standard excess returns, with average adjusted R^2 around 0.923.

Table 7
Continued

D. Portfolios sorted by equity momentum						
	Low	2	3	4	High	High - Low
Avg. $r_{i,t} - r_{f,t}$	0.171 (1.52)	0.354 (4.44)	0.385 (5.02)	0.443 (5.56)	0.563 (6.30)	0.392 (5.02)
α	-0.107 (-1.20)	0.114 (1.58)	0.148 (2.13)	0.207 (2.75)	0.341 (3.97)	0.448 (6.38)
Adj. R^2	.360	.461	.496	.502	.443	.217
Mean $ \alpha $	0.183					
CAPM α	-0.010 (-0.10)	0.262 (3.21)	0.306 (3.81)	0.371 (4.35)	0.483 (5.26)	0.493 (6.65)
CAPM adj. R^2	.279	.124	.098	.078	.102	.181
CAPM mean $ \alpha $	0.286					

This table presents time-series regressions of the excess returns of corporate bond portfolios sorted by four characteristics on the duration-matched Treasury and dividend risk components of the S&P 500 Index excess return:

$$r_{i,t} - r_{f,t} = \alpha + \beta_{TSY}(r_{TSY,t} - r_{f,t}) + \beta_{DurAdj}(r_{SP500,t} - r_{TSY,t}) + \epsilon_{i,t}.$$

The sample consists of 419 monthly observations from January 1986 to December 2020 for credit spread, downside risk, and equity-momentum portfolios while it is 221 months from August 2002 to December 2020 for illiquidity-sorted portfolios. Value-weighted returns are computed using bond-level quote data from Lehman Brothers and Bank of America Merrill Lynch. Excess returns are computed by subtracting the 1-month Treasury-bill rate. t -statistics based on Newey and West (1987) standard errors with 12 lags are reported in parentheses.

Table 9
Model comparison

A. Summary statistics of factors											
Model	CAPM		Dur.-adj. CAPM		Dur.-adj. bond CAPM		TERM and DEF factors		Bai et al. 2019 model		
	SP500	Trsy	DurAdj	Trsy	DurAdj	TERM	DEF	Mkt	CRF	DRF	LRP
Mean	0.793	0.813	-0.020	0.279	0.138	0.510	0.048	0.473	0.385	0.646	0.345
Std	(2.52)	(2.54)	(-0.04)	(0.03)	(0.90)	(2.68)	(0.03)	(2.96)	(1.31)	(2.65)	(2.37)
SD (normalized)	4.212	5.010	7.605	1.387	1.966	3.590	2.171	1.807	3.196	3.237	1.372
SP (annualized)	0.652	0.561	-0.009	0.697	0.244	0.554	0.077	0.879	0.417	0.691	0.870
CAPM α	1.148	-1.148	0.363	-0.129	0.665	-0.185	0.285	-0.006	0.355	0.266	
t -stat	(3.22)	(-3.22)	(3.73)	(-1.35)	(1.12)	(-1.63)	(2.34)	(-0.03)	(1.64)	(2.12)	
DCAPM α	0.677	-0.057	-0.015	-0.008	-0.083	0.113	-0.278	0.215	0.233		
t -stat	(1.37)	(-0.12)	(-0.09)	(-0.73)	(0.79)	(-1.38)	(0.79)	(1.38)	(0.79)	(1.44)	
B. Squared Sharpe ratio and pairwise model comparison											
Model	CAPM		Dur.-adj. CAPM		Dur.-adj. bond CAPM		TERM and DEF		Bai et al. 2019 model		
	SP500	Trsy	DurAdj	Trsy	DurAdj	TERM	DEF	Mkt	CRF	DRF	LRP
Adj. SP^2	0.030	0.084	0.086	0.026	0.066			0.026			
Mean DCAPM $ \alpha $			0.045	0.210	0.089			0.210			
GRS p -value			.233	.757	.089			.757			
Pairwise model comparison based on differences in adjusted SP^2											
CAPM			-0.054	-0.036	0.004			0.004			-0.037
Decomposed CAPM			1.181	1.791	1.738			1.966			1.389
TERM and DEF			0.018	0.058	0.017			0.017			0.017
			1.678	1.881	1.678			1.881			1.741
				0.040	0.001			0.040			0.001
				1.115	1.931			1.115			1.798

This table presents the summary statistics of asset pricing factors including the CAPM, duration-adjusted CAPM, duration-adjusted bond CAPM, TERM and DEF factors of Fama and French (1993), and the four factors of Bai, Ball, and Wei (2019). The sample consists of 197 monthly observations from August 2004 to December 2020. For each factor, we calculate the mean, standard deviation, and annualized Sharpe ratio (SP). We also report the intercept of the regression of each factor on the market factor (CAPM α) and on the two factors of the duration-adjusted CAPM (DCAPM α). Values in parentheses are t -statistics using Newey and West (1987) standard errors with 12 lags. For each model, we calculate the adjusted maximum squared monthly Sharpe ratio following Barillas et al. (2020). Mean DCAPM $|\alpha|$ is the average absolute alpha across the duration-adjusted CAPM across the factors in each model. GRS p -value is the p -value of the Gibbons, Ross, and Shanken (1989) test using the factors in each model as test assets. Values in brackets are the p -value of the differences in the squared Sharpe ratio.

- That is even higher than the Bai–Bali–Wen model’s average adjusted R^2 of about 0.876 in panel A.
- The Bai model has slightly smaller average absolute alphas in that panel, but its advantage is much less clear once attention turns to duration-adjusted returns.
- In Table 9, the simple duration-adjusted CAPM delivers the highest bias-adjusted squared Sharpe ratio among the models compared.

Economic message. The paper’s duration-based approach is competitive even against much richer bond-factor models. The main lesson is not that all existing bond factors are useless; it is that duration adjustment is first-order and should come before layering on additional factors.

6 Short summary

- The paper argues that standard corporate bond excess returns are the wrong object for measuring credit premia in a falling-rate era.
- It decomposes each bond’s realized return into a duration-matched Treasury return and a duration-adjusted residual.
- Most realized excess returns—especially for investment-grade bonds—turn out to be Treasury-duration windfalls rather than compensation for credit/liquidity risk.
- The resulting duration-adjusted risk premia are small and statistically weak.
- The standard CAPM fails badly on unadjusted corporate bond excess returns but works much better on duration-adjusted returns.
- A two-factor duration-adjusted CAPM and a duration-adjusted bond CAPM both perform well, and much of the apparent value of richer bond-factor models seems to come from loading on duration.
- Duration adjustment matters for credit-spread, downside-risk, and illiquidity portfolios, though not for bond momentum.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a large part of realized corporate bond excess returns over the last four decades reflects the return on long-duration government bonds, not compensation for bearing corporate credit and liquidity risk. Once corporate bond returns are measured relative to a duration-matched Treasury benchmark, investment-grade premia become very small and the CAPM’s apparent failure largely disappears.

7.2 Economic intuition

In a secularly declining yield environment, any long-duration asset enjoys capital gains. Investment-grade corporate bonds are partly risky corporate claims, but they are also partly long-duration bond positions. If researchers subtract only the Treasury-bill rate, they mistakenly treat the long-term Treasury component as a corporate-risk premium. The paper's duration adjustment peels those two pieces apart.

7.3 Takeaway

The paper's deepest message is methodological: when discount-rate environments are nonstationary, *excess returns relative to cash can be a misleading object for asset pricing*. For corporate bonds, the right benchmark is not cash but a safe asset with the same duration exposure. Once that benchmark is used, corporate bond risk premia and pricing “anomalies” look much smaller and much less mysterious.